SRI RAMACHANDRA
INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Category - I Deemed to be University) Porur, Chennai
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

Healthcare Centre Network Infrastructure

CSE23CL301 - Computer Networks Laboratory

PROJECT REPORT

Submitted by

Suba Lakshmi S

E0124032

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

(Artificial Intelligence and Machine Learning)

Sri Ramachandra Faculty of Engineering and Technology,

Sri Ramachandra Institute of Higher Education and Research, Porur, Chennai -600116

OCTOBER , 2026

SRI RAMACHANDRA
INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Category - I Deemed to be University) Porur, Chennai
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

BONAFIDE CERTIFICATE

I, Suba Lakshmi .S - E0124032 hereby declare that the Project Report on **"Healthcare Centre Network Infrastructure "** is submitted by me under the guidance of **Prof Mr.G.THANIGAIVEL** for **CSE23CL301 -Computer Networks Laboratory**, at Sri Ramachandra Faculty of Engineering and Technology, in partial fulfilment of Sem 5, Third Year B.Tech Degree course in the Department of Computer Science and Engineering during the academic year 2025-2026

Prof Mr.G.THANIGAIVEL

Course faculty,

Assistant Professor,

Sri Ramachandra Faculty of Engineering and Technology,

Sri Ramachandra Institute of Higher Education and Research,

Porur, Chennai, Tamil Nadu.

Submitted for the Computer Networks Laboratory project presentation held at Sri Ramachandra Faculty of Engineering and Technology on _____

Internal Examiner **External Examiner**

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	1
	LIST OF FIGURES	2
1	INTRODUCTION 1.1 Project Overview 1.2 Objectives 1.3 Scope	3
2	SYSTEM REQUIREMENTS 2.1 Hardware Requirements 2.2 Software Requirements	6
3	DESIGN OF PROPOSED SYSTEM	8
4	ALGORITHM	12
5	IMPLEMENTATION	14
6	RESULTS	20
7	CONCLUSION 7.1 Summary 7.2 Future Enhancement	21

ABSTRACT

The Healthcare Centre Network Infrastructure project focuses on designing and implementing a reliable, organized, and secure computer network for a healthcare centre using Cisco Packet Tracer. In a healthcare environment, different departments need to communicate with one another and access shared resources such as servers, applications, and network services. A properly designed network helps improve communication, resource sharing, and the overall efficiency of hospital operations.

The proposed network connects major departments, including Administration, Reception and Registration, Consultation, Diagnostics and Radiology, Laboratory, Pharmacy, IT, Server Room, Guest Wi-Fi, and CCTV. Virtual Local Area Networks (VLANs) are used to separate departmental traffic and organize the network. Two multilayer core switches provide inter-VLAN routing, while Hot Standby Router Protocol (HSRP) is configured to support gateway redundancy. EtherChannel is implemented between the core switches to combine physical links into a logical connection and improve link capacity.

The infrastructure also includes an Adaptive Security Appliance (ASA) firewall, internet-edge and branch routers, and dedicated servers for services such as web hosting, DNS, DHCP, applications, databases, and file storage. A sample healthcare information website named CarePoint Healthcare Centre is hosted to demonstrate web services within the network.

The project is designed and tested in Cisco Packet Tracer to understand practical networking concepts, device configuration, IP addressing, routing, VLAN segmentation, and network troubleshooting. The completed simulation provides a foundation for developing a more secure, scalable, and efficient healthcare network in the future.

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
Fig 3.1	Overall Healthcare Centre Network Topology	9
Fig 5.1	Department-Wise VLAN Architecture	14
Fig 5.2	Core Switches and EtherChannel Connection	15
Fig 5.3	HSRP Configuration on Core Switches	15
Fig 5.4	Server Room and Internal Server Connections	16
Fig 5.5	ASA Firewall and Network Transit Connection	16
Fig 5.6	Router and Branch Network Topology	17
Fig 5.7	DNS Service Configuration	18
Fig 5.8	CarePoint Healthcare Centre Website Configuration	19
Fig 5.9	Network Connectivity Testing Using Ping	19

CHAPTER 1

INTRODUCTION

1.1 Project Overview

A healthcare centre depends on the continuous exchange of information between its departments. Administration handles office operations, reception manages patient registration, doctors provide consultations, and laboratory and pharmacy departments support patient care. These departments often require access to common network resources while maintaining appropriate separation between their systems.

The Healthcare Centre Network Infrastructure project aims to create a structured network that connects the different departments of a healthcare centre using Cisco Packet Tracer. The network is designed to provide communication between authorized devices, enable access to centralized servers, and demonstrate important networking concepts in a simulated environment.

The network contains separate departmental switches connected to two central multilayer switches, CORE-SW1 and CORE-SW2. VLANs are used to logically separate departments, while inter-VLAN routing enables communication between different network segments when permitted by the configuration. HSRP is configured to provide redundant default gateways for supported VLANs, and EtherChannel combines two physical links between the core switches into a single logical link.

The proposed infrastructure also includes an ASA firewall, routers for internet-edge and branch connectivity, and a centralized server network. The servers are intended to support web, DNS, DHCP, application, database, and file services. A CarePoint Healthcare Centre website is included as a practical example of a web service hosted within the network.

Cisco Packet Tracer makes it possible to configure devices, observe network behaviour, and test connectivity without requiring physical networking equipment. This project provides practical experience in network design, device configuration, troubleshooting, and infrastructure planning.

1.2 Objectives

The main objective of this project is to design a structured and manageable network infrastructure for a healthcare centre using Cisco Packet Tracer.

The specific objectives are:

1. To design a network that connects the major departments of a healthcare centre.
2. To divide the network into separate VLANs according to departmental requirements.
3. To configure inter-VLAN routing using multilayer switches.
4. To implement HSRP for default-gateway redundancy on supported VLANs.
5. To configure EtherChannel between the two core switches to combine physical links.
6. To establish a centralized server network for shared network services.
7. To incorporate an ASA firewall into the proposed network architecture.
8. To configure routers for communication between the main network and branch network.
9. To demonstrate web hosting and DNS configuration through the CarePoint Healthcare Centre website.
10. To test network connectivity and identify configuration issues through practical troubleshooting.

1.3 Scope

The scope of this project covers the design, configuration, and simulation of a healthcare centre network using Cisco Packet Tracer. It focuses on the practical implementation of basic and intermediate networking concepts in a departmental environment.

The project includes departmental VLANs, multilayer switching, inter-VLAN routing, HSRP, EtherChannel, server connectivity, router connections, and an ASA firewall. It also includes the configuration of network services and a sample website to demonstrate how internal users can access hosted resources.

The design separates the Administration, Reception and Registration, Consultation, Diagnostics and Radiology, Laboratory, Pharmacy, IT, Server Room, Guest Wi-Fi, and CCTV network segments. This structure makes the network easier to manage and provides a basis for applying access-control policies between departments.

The project is implemented as a simulation rather than a production healthcare network. Features such as complete internet connectivity, firewall security policies, DMZ isolation, ISP failover, encrypted site-to-site VPN connectivity, and end-to-end security enforcement require additional configuration and testing before they can be considered fully operational.

CHAPTER 2

SYSTEM REQUIREMENTS

2.1 Hardware Requirements

The project can be developed using a personal computer or laptop capable of running Cisco Packet Tracer.

Component	Requirement
Processor	Intel Core i3 or equivalent and above
RAM	Minimum 4 GB; 8 GB or above recommended
Storage	At least 2 GB of free space for the application and project files
Display	Standard monitor or laptop display
Input devices	Keyboard and mouse
Network devices in simulation	Routers, multilayer switches, access switches, ASA firewall, PCs, servers, and access points

The listed hardware requirements refer to the computer used to develop and run the simulation. The routers, switches, servers, and other network devices are represented virtually inside Cisco Packet Tracer.

2.2 Software Requirements

The project uses networking simulation software and network configuration tools to design, implement, and verify the proposed enterprise network.

The following software components are required:

Software	Purpose
Cisco Packet Tracer	Designing, configuring, and testing the network topology
Windows operating system	Running the simulation software
HTML	Creating the healthcare centre website
Web browser in Packet Tracer	Accessing and testing the simulated web service
Cisco IOS CLI in Packet Tracer	Configuring routers and multilayer switches
ASA command-line interface	Configuring and inspecting the simulated firewall

Cisco Packet Tracer provides a virtual environment in which network devices can be connected and configured without requiring all the physical equipment. Its command-line interface allows the implementation of routing, VLANs, redundancy, and other networking concepts.

HTML is used to create the CarePoint Healthcare Centre informational website. The website is hosted through the simulated server environment.

CHAPTER 3

DESIGN OF PROPOSED SYSTEM

The proposed system uses a hierarchical network design in which departmental devices connect to access switches, access switches connect to the core layer, and centralized services are connected through the server network.

The design is intended to simplify network management, separate departmental traffic, and improve the availability of essential network connections.

3.1 Overall Network Architecture

The healthcare centre network consists of the following major components:

1. Departmental Access Layer

Each department has a dedicated access switch connecting computers, printers, laptops, and wireless access points. This arrangement allows devices to connect to the network according to their department.

2. Core Network Layer

CORE-SW1 and CORE-SW2 act as the central multilayer switches. They provide Layer 3 routing for configured VLANs and connect the departmental networks to the server network and other network segments.

3. Server Network

The server network contains dedicated servers for different services, including application support, database services, DHCP, file storage, and internal web hosting. Centralizing these services makes them easier to manage.

4. Firewall and External Connectivity

An ASA firewall is included in the proposed architecture to represent the boundary between the internal healthcare network and external connectivity. Its security policies and routing must be configured according to the intended network paths.

5. DMZ Network

The topology includes DMZ-WEB, DMZ-DNS, and DMZ-EMAIL servers. A properly configured DMZ should use a separate network segment and firewall policies that restrict access to internal systems.

6. Branch Network

Routers connect the main network and branch network. This demonstrates how a healthcare organization can extend its network to another location.

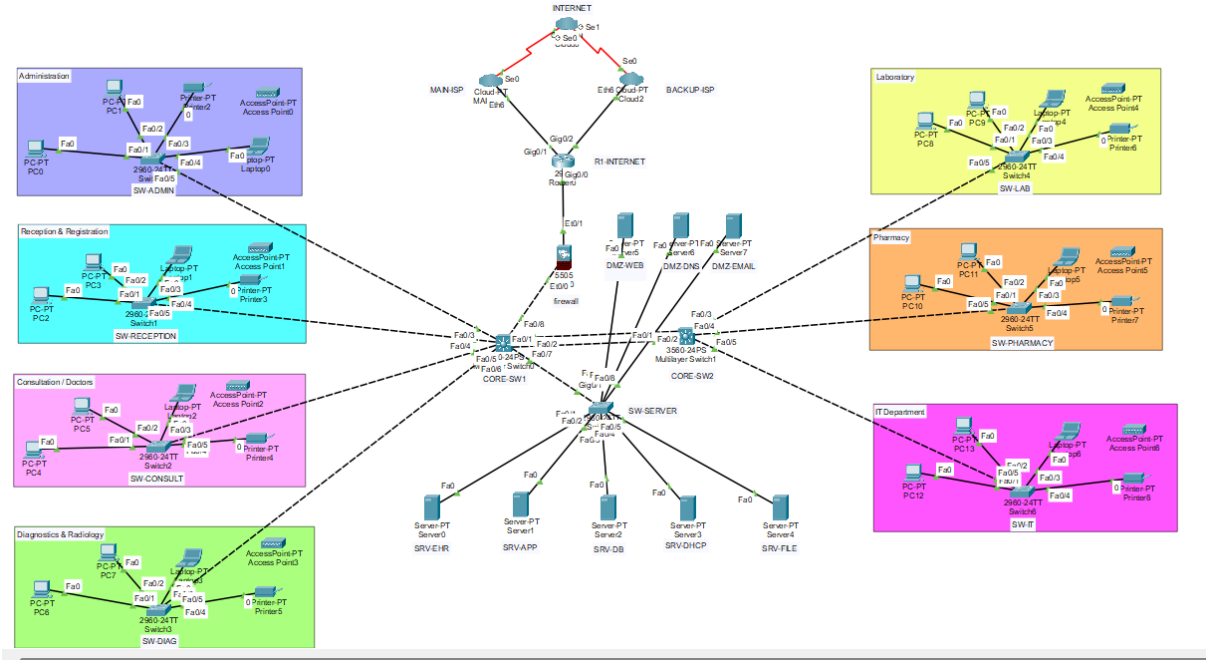


Figure 3.1: Overall Healthcare Centre Network Topology

3.2 VLAN Design

VLANs are used to logically separate departmental traffic within the network. Each VLAN is assigned a unique network address.

VLAN ID	VLAN Name	Network Address
10	ADMINISTRATION	192.168.10.0/24
20	RECEPTION	192.168.20.0/24
30	CONSULTATION	192.168.30.0/24
40	DIAGNOSTICS	192.168.40.0/24
50	LABORATORY	192.168.50.0/24
60	PHARMACY	192.168.60.0/24
70	SERVERS	192.168.70.0/24
80	IT	192.168.80.0/24
90	GUEST_WIFI	192.168.90.0/24
100	CCTV	192.168.100.0/24
110	FIREWALL_TRANSIT	10.10.30.0/30

VLANs reduce the need to place every device in a single broadcast domain. They also provide a foundation for applying access-control policies between departments. VLAN separation alone does not guarantee security; routing rules and access controls are also required.

3.3 IP Addressing and Routing

Each departmental VLAN uses its own IP subnet. The multilayer switches provide the Layer 3 interfaces needed for inter-VLAN routing.

CORE-SW1 is configured with gateway addresses for the departmental VLANs, while CORE-SW2 provides corresponding addresses for the configured HSRP groups. A virtual IP address is used as the default gateway for supported VLANs.

The server network uses the 192.168.70.0/24 subnet. The firewall transit network uses 10.10.30.0/30 in the current configuration. Router-to-router links use separate subnets to support communication between the main network and the branch network.

Static routing is used in the simulated environment to define paths between network segments. Correct routing depends on the configured next-hop addresses and return routes.

3.4 Redundancy and Link Aggregation

HSRP is used to provide gateway redundancy. One core switch operates as the active gateway for a configured VLAN, while the other is available as the standby gateway. If the active gateway becomes unavailable, the standby switch can take over, provided the HSRP configuration and connectivity are correct.

EtherChannel combines multiple physical links into one logical connection. In this project, FastEthernet0/1 and FastEthernet0/2 are bundled into Port-channel1 between the core switches. The show etherchannel summary command is used to inspect the status of the logical link and its member interfaces.

These features improve the resilience and availability of the network, although full failover behaviour should be confirmed through controlled testing.

3.5 Server and Website Design

The server layer contains dedicated systems for internal services. The proposed functions include employee or healthcare-related application support, database services, DHCP, file storage, and web hosting.

The CarePoint Healthcare Centre website is created using HTML. It includes a home section, department descriptions, healthcare services, sample doctor profiles, appointment instructions, and contact information. The website is informational and does not provide actual medical services or store appointment requests.

DNS is intended to map the hostname healthcare.local to the IP address of the selected web server. For the hostname to work, the DNS record, client DNS configuration, server IP address, and HTTP service must all be correct.

3.6 Firewall and DMZ Design

The ASA firewall represents the security boundary between internal and external network segments. The design includes an outside interface and an inside interface, with DMZ services intended to be placed in a separate security zone.

The internal network should not be directly exposed to public-facing services. Firewall rules should permit only required traffic, such as HTTP or HTTPS to the web server, and restrict unnecessary access to internal servers.

The DMZ arrangement shown in the topology represents the intended architecture. Separate DMZ addressing, correct firewall interfaces, routing, and access-control rules are required to achieve actual isolation.

CHAPTER 4

ALGORITHM

Algorithm for Enterprise Network Implementation

The network operation can be described as a sequence of steps that explains how devices connect, obtain network settings, communicate with servers, and access the healthcare website.

Step 1: Initialize the Network

Start the network simulation and verify that the configured routers, switches, firewall, servers, and client devices are powered on and connected.

Step 2: Configure VLANs

Create the required VLANs on the core and access switches. Assign the appropriate access ports to their departmental VLANs.

Step 3: Configure IP Addressing

Assign IP addresses and subnet masks to the Layer 3 interfaces. Configure the required gateway addresses and routing interfaces.

Step 4: Configure Inter-VLAN Routing

Enable IP routing on the multilayer switches and configure the VLAN interfaces. This allows communication between different subnets according to the routing configuration.

Step 5: Configure HSRP

Configure HSRP groups and virtual IP addresses for the supported departmental VLANs. Set the intended active and standby roles.

Step 6: Configure EtherChannel

Bundle the selected physical interfaces between CORE-SW1 and CORE-SW2. Verify that the port-channel and member interfaces are operational.

Step 7: Configure DHCP

Configure the DHCP server with the required address pools, subnet masks, default gateways, and DNS server information. Configure DHCP relay addresses on the relevant VLAN interfaces when the DHCP server is located in another subnet.

Step 8: Configure the Firewall and Routing

Assign the firewall interfaces to the intended security zones. Configure the required routes and security policies for communication between internal, external, and DMZ networks.

Step 9: Configure the Servers

Assign the correct IP addresses and gateways to the internal servers. Enable the required services, including HTTP, DNS, and DHCP, according to each server's role.

Step 10: Configure the Website and DNS Record

Store the HTML website in the web server's index.html file. Create a DNS A record mapping healthcare.local to the actual IP address of that web server.

Step 11: Test Connectivity

Use ping commands to verify reachability between clients, gateways, and servers. Check routing tables, VLAN membership, HSRP status, and EtherChannel status.

Step 12: Test Web Access

Open the web browser on a client PC and access the website using the server IP address. After confirming direct IP access, test the hostname healthcare.local.

Step 13: Save the Configuration

Save the Packet Tracer project and relevant device configurations after completing the required checks.

CHAPTER 5

IMPLEMENTATION

The project is implemented in Cisco Packet Tracer by creating the network topology, configuring devices, assigning IP addresses, and testing connectivity between the different network components.

5.1 Network Topology Creation

The topology is created using routers, multilayer switches, access switches, an ASA firewall, servers, PCs, printers, and wireless access points. The devices are arranged according to their functions and departmental locations.

Separate access switches are used for Administration, Reception and Registration, Consultation and Doctors, Diagnostics and Radiology, Laboratory, Pharmacy, and IT. Guest Wi-Fi and CCTV are also represented in the design.

The access switches are connected to the core network, while the server switch provides connectivity to the internal servers. Routers are used to represent external and branch network connectivity.

```
CORE-SW1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   ADMINISTRATION                   active
20   RECEPTION                        active
30   CONSULTATION                     active
40   DIAGNOSTICS                      active
50   LABORATORY                       active
60   PHARMACY                         active
70   SERVERS                          active
80   IT                               active
90   GUEST_WIFI                       active
100  CCTV                             active
110  FIREWALL_TRANSIT                 active    Fa0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
```

FIG 5.1: Department-Wise VLAN Architecture

5.2 VLAN Configuration

VLANs are created on the core switches using their VLAN IDs and names. Departmental access ports are assigned to the appropriate VLANs.

The core switches are configured with VLAN interfaces to provide routing between the departmental networks. The configured IP addresses serve as the Layer 3 gateway addresses for the corresponding subnets.

The show vlan brief command is used to verify VLAN creation and access-port membership. The show ip interface brief command is used to inspect the status of interfaces.

```
CORE-SW1>show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------

1      Po1(SU)           -      Fa0/1(P) Fa0/2(P)
CORE-SW1>
```

FIG 5.2: Core Switches and EtherChannel Connection

5.3 HSRP Configuration

HSRP is configured on the core switches for the supported departmental VLANs. Each HSRP group uses a virtual IP address that client devices can use as their default gateway.

The active and standby roles are selected through the HSRP configuration, including priority and preemption settings where required. The show standby brief command is used to inspect the current HSRP roles and virtual IP addresses.

This configuration demonstrates how a redundant gateway arrangement can reduce dependence on a single Layer 3 switch.

```
CORE-SW1>enable
CORE-SW1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           192.168.10.2    192.168.10.254
Vl20        20   90  P Standby  192.168.20.2    local           192.168.20.254
Vl30        30   110 P Active   local           192.168.30.2    192.168.30.254
Vl40        40   110 P Active   local           192.168.40.2    192.168.40.254
Vl50        50   110 P Active   local           192.168.50.2    192.168.50.254
Vl60        60   110 P Active   local           192.168.60.2    192.168.60.254
Vl80        80   110 P Active   local           192.168.80.2    192.168.80.254
Vl90        90   110 P Active   local           192.168.90.2    192.168.90.254
Vl100       100  110 P Active   local           192.168.100.2   192.168.100.254
CORE-SW1#
```

Fig 5.3: HSRP Configuration on Core Switches

5.4 EtherChannel Configuration

EtherChannel is configured between CORE-SW1 and CORE-SW2 using FastEthernet0/1 and FastEthernet0/2.

The physical interfaces are combined into Port-channel1. This creates a logical link between the switches while allowing the physical member links to contribute to the connection.

The show etherchannel summary command is used to check the result. In the observed configuration, Port-channel1 is reported as SU, and both FastEthernet0/1 and FastEthernet0/2 are marked as bundled ports (P).

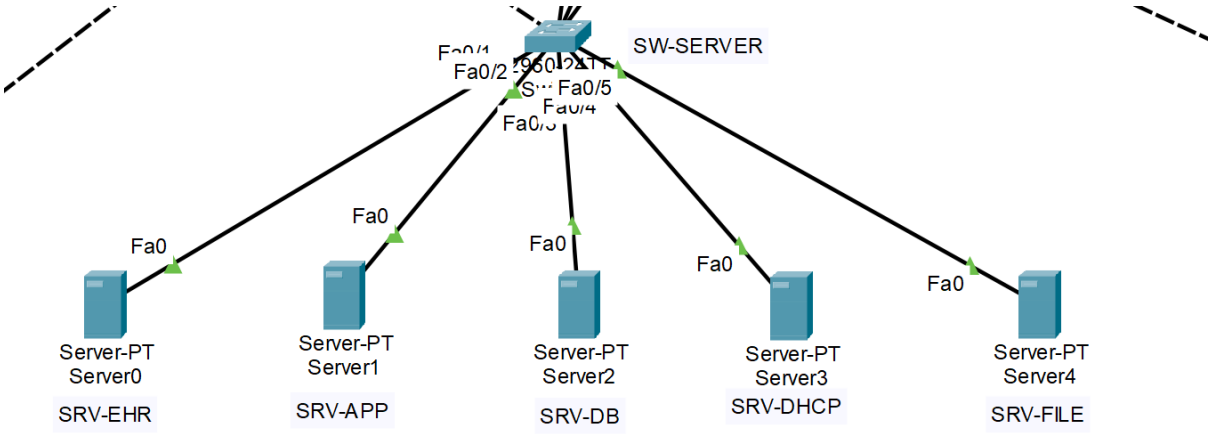


Fig 5.4: Server Room and Internal Server Connections

5.5 DHCP and Server Configuration

A dedicated DHCP server is included in the server network. DHCP relay configuration is used on the relevant VLAN interfaces so that client requests can reach the server across subnet boundaries.

The server infrastructure also includes systems intended for application, database, file, and web services. Each server must have an appropriate IP address, subnet mask, default gateway, and service configuration.

Client devices can be configured to obtain their network settings automatically. The assigned IP address, gateway, and DNS server should be checked to confirm that DHCP is operating correctly.

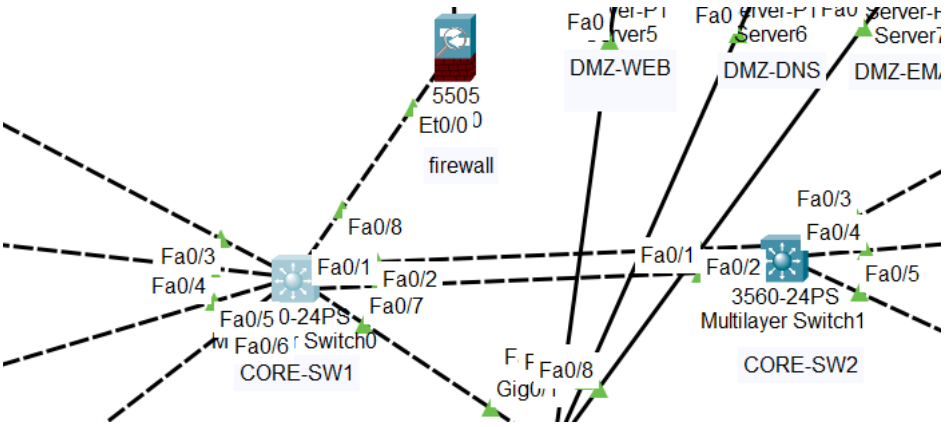


FIG 5.5: ASA Firewall and Network Transit Connection

5.6 Firewall and DMZ Implementation

The ASA firewall is included in the network topology. The configured outside interface uses the 10.10.30.2 address, while the inside interface is configured with 192.168.1.1 in the current configuration.

The core switch has a firewall transit VLAN using the 10.10.30.0/30 subnet. Connectivity between CORE-SW1 and the ASA outside address was verified successfully using a ping test.

For the final security design, the inside interface must be connected to a suitable internal transit network, and the routes must match the actual physical and logical topology. The DMZ servers must also be placed in a separate network segment with suitable firewall rules. These elements require separate verification before claiming full internet connectivity or DMZ isolation.

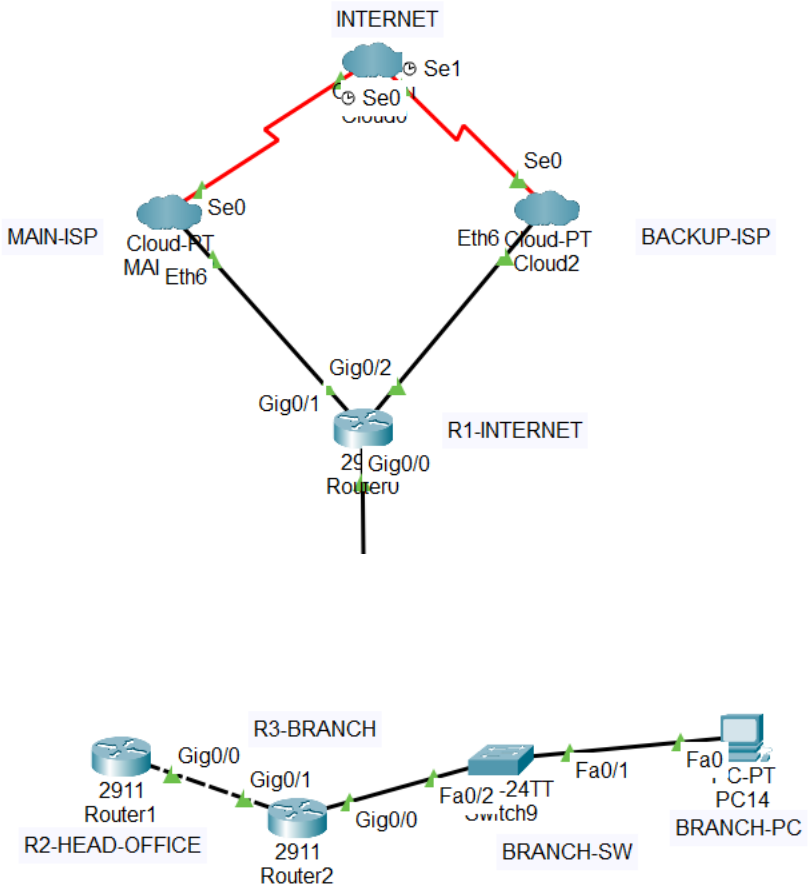


FIG 5.6: Router and Branch Network Topology

5.7 Website Implementation

The CarePoint Healthcare Centre website is developed using HTML and hosted through the simulated server environment.

The page contains a header, navigation links, a welcome section, department information, healthcare services, sample doctor profiles, appointment instructions, and contact details. The layout uses colours and tables to present information in a structured format.

The website is intended to be accessed from client PCs through the simulated network. The HTTP service must be enabled, and the server's IP address must be reachable from the client.

The DNS service is configured with the hostname healthcare.local. To complete the website access test, the DNS record must point to the actual web server IP address, and the client must use the correct DNS server.

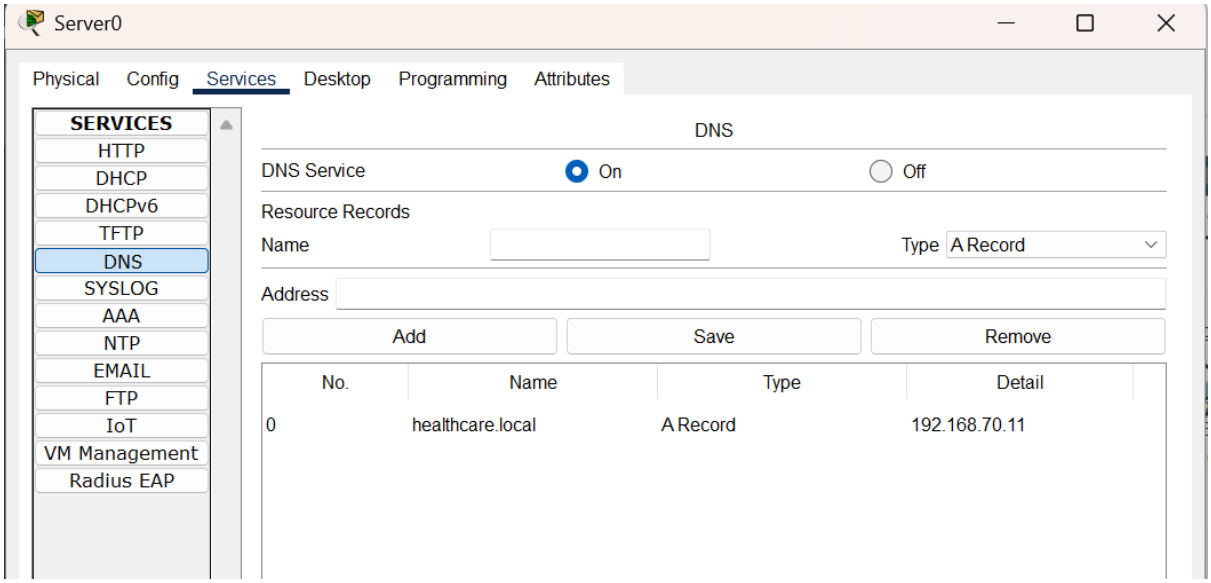


FIG 5.7: DNS Service Configuration

5.8 Network Testing and Troubleshooting

Network testing is performed using the command-line interfaces of the core switches, routers, and firewall, along with the desktop tools on client PCs.

The following commands are used during verification:

- show vlan brief — checks VLAN creation and port membership.
- show ip interface brief — checks interface addresses and operational status.
- show standby brief — checks HSRP roles and virtual IP addresses.
- show etherchannel summary — checks EtherChannel status.
- show ip route — checks the routing table on a router or multilayer switch.
- ping <IP-address> — tests IP connectivity to another device.

When the website does not open, the troubleshooting process checks the client IP configuration, server IP address, gateway, DNS record, HTTP service, and routing path. Testing the web server directly by IP address helps distinguish a DNS issue from an HTTP or network connectivity issue.

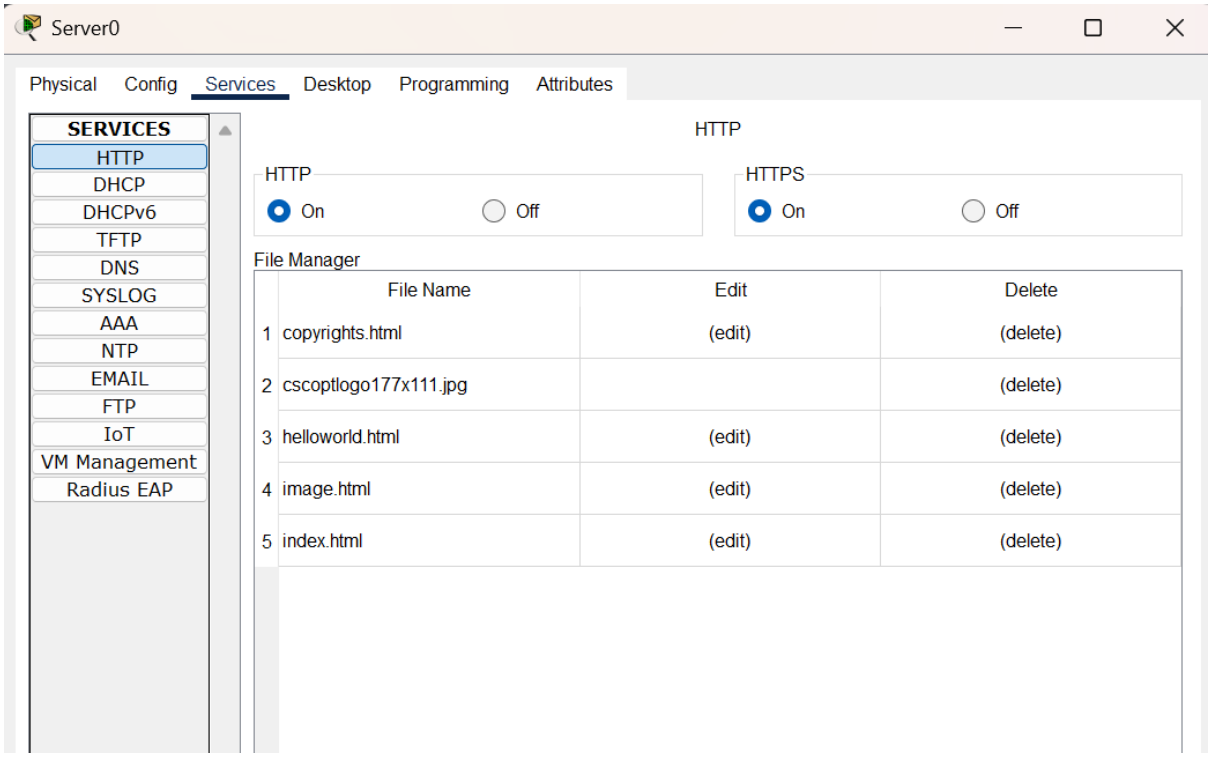


FIG 5.8: CarePoint Healthcare Centre Website Configuration

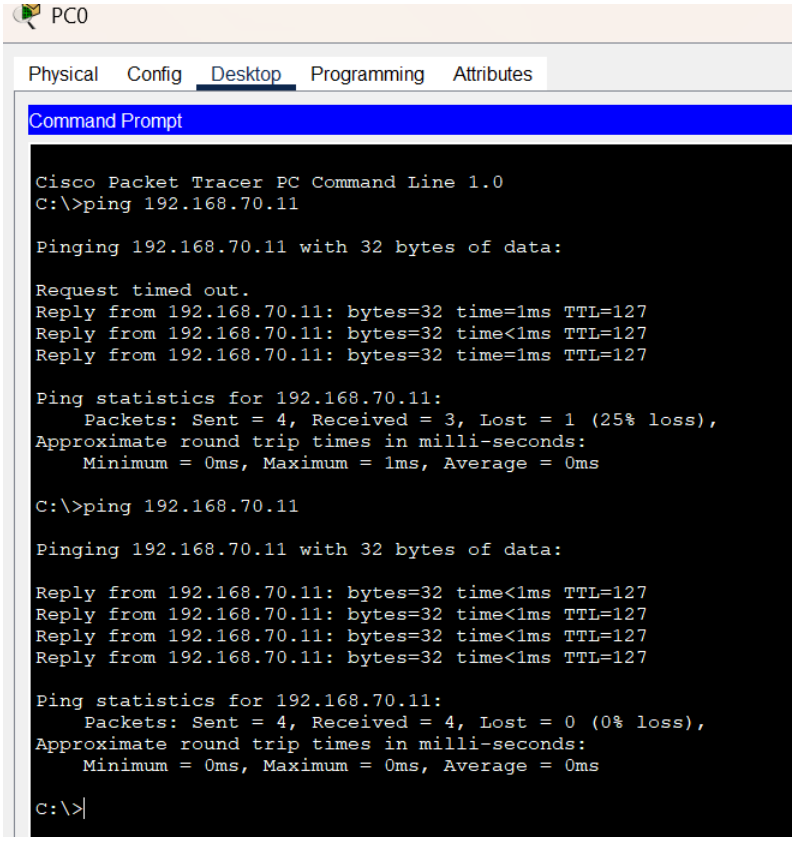


Fig 5.9: Network Connectivity Testing Using Ping

CHAPTER 6

RESULTS

The Healthcare Centre Network Infrastructure project was developed in Cisco Packet Tracer to demonstrate a structured network design for a healthcare environment.

The VLAN configuration was inspected using the show vlan brief command. The required departmental VLANs and the firewall transit VLAN were present in the core switch configuration.

The HSRP configuration was checked using show standby brief. The output showed active and standby gateway roles for the configured VLANs, demonstrating the intended gateway redundancy arrangement.

EtherChannel was verified using show ether channel summary. Port-channel1 was shown as operational, with FastEthernet0/1 and FastEthernet0/2 participating in the channel.

Connectivity between CORE-SW1 and the ASA outside interface was tested successfully. The ping to 10.10.30.2 returned five successful replies, indicating reachability across the configured transit link.

The internal web server address 192.168.70.11 was also reachable from CORE-SW1. The initial ping returned one timeout followed by successful replies, and a subsequent ping completed with 100% success. This demonstrated IP reachability between the core switch and that server address.

The CarePoint Healthcare Centre website was prepared using HTML, and the server's HTTP service was shown as enabled. However, the browser tests produced hostname-resolution and request-timeout errors during troubleshooting. Therefore, successful end-to-end website access through both the server IP address and healthcare local still requires final verification.

The project demonstrates the configuration of departmental VLANs, gateway redundancy, link aggregation, server connectivity, and a simulated firewall boundary. Further validation is required for complete DNS resolution, website access, internet routing, NAT, and DMZ isolation.

CHAPTER 7

CONCLUSION

The Healthcare Centre Network Infrastructure project demonstrates how a structured computer network can be designed for the different departments of a healthcare centre. Cisco Packet Tracer was used to build the topology and configure VLANs, multilayer switches, HSRP, EtherChannel, routers, servers, and an ASA firewall.

The project provides practical experience in network segmentation, IP addressing, routing, gateway redundancy, link aggregation, and connectivity troubleshooting. It also introduces the role of centralized servers and an informational website in a healthcare network environment.

The observed test results confirmed that the core-to-firewall transit connection was reachable, the core switch could reach the internal web server address, the configured HSRP roles were visible, and the EtherChannel was operational. At the same time, the website browser tests highlighted the importance of verifying DNS records, HTTP services, client settings, and routing paths before declaring end-to-end service availability.

Overall, the project provides a useful foundation for understanding enterprise network design and the requirements of a healthcare centre network. With additional testing and security configuration, the design can be extended into a more complete and reliable simulated healthcare network.

7.1 Summary

The project can be enhanced in several ways to make the network more secure, scalable, and suitable for a larger healthcare environment.

1. Complete Firewall Security Policies

Configure and test firewall access-control rules to restrict unauthorized communication between the internal network, external networks, and DMZ.

2. Complete Internet Connectivity

Configure the required ISP addressing, routing, NAT, and return paths to enable verified external connectivity through the intended primary and backup links.

3. Improve DMZ Isolation

Assign the DMZ servers to a separate subnet and security zone. Apply firewall rules that permit only necessary access to public-facing services.

4. Complete DNS and Web Server Testing

Verify the web server's actual IP address, HTTP service, DNS record, and client DNS settings. Confirm that both direct IP access and healthcare.local work successfully.

5. Strengthen Network Access Control

Implement suitable access-control lists to restrict communication between departments. Guest Wi-Fi should be isolated from sensitive server networks, and CCTV devices should have limited access to required services.

6. Add Network Monitoring

Introduce centralized logging, monitoring, and alerting to identify interface failures, unusual traffic, and service interruptions.

7. Improve High Availability

Test HSRP failover and EtherChannel member-link failure. Additional redundancy can be considered for critical network components.

8. Develop Healthcare Applications

The informational website can be extended into a prototype healthcare application with patient registration, appointment management, and department information. Any real patient data would require appropriate privacy, security, authentication, and access-control safeguards.

9. Expand the Network

The design can be extended to additional departments, hospital buildings, and branch locations using structured IP addressing and suitable routing protocols.

These enhancements would help move the project from a basic network simulation toward a more complete enterprise healthcare network design.